

STA326: Data Science Practice

Wei Hongxin
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Quiz Time



* 1. 以下哪个现象最可能表明模型发生了过拟合?

[5分]

- ☐ 训练集准确率90%，测试集准确率85%
 - ☒ 训练集准确率98%，测试集准确率60% (正确答案)
 - ☐ 训练集准确率50%，测试集准确率45%
 - ☐ 训练集准确率70%，测试集准确率80%
-

* 2. BN的主要作用是什么?

[5分]

- ☐ 减少模型参数量
- ☐ 标准化输入数据的分布
- ☒ 减少内部协变量偏移 (正确答案)
- ☐ 增加模型的非线性

* 3. LN特别适合以下哪种场景?

[5分]

- ☐ 批量大小较大
 - ☒ 序列数据处理 (正确答案)
 - ☐ 图像分类任务
 - ☐ 以上都是
-

* 4. 在正则化中, 增大正则强度会如何影响模型?

[5分]

- ☐ 模型复杂度增加
- ☒ 模型复杂度减少 (正确答案)
- ☐ 训练误差减少
- ☐ 测试误差增加

* 5. BN在测试阶段使用的全局统计是如何得到的?

[5分]

- ☐ 从训练集的每个批量实时计算
 - ☒ 从训练过程中累积的移动平均值 (正确答案)
 - ☐ 从测试集计算
 - ☐ 从验证集计算
-

* 6. Dropout的本质可以看作是对模型的哪种操作?

[5分]

- ☐ 增加模型深度
- ☒ 集成多个子模型 (正确答案)
- ☐ 减少参数量
- ☐ 提高学习率

* 7. 早停 (Early Stopping) 是通过什么机制减少过拟合的?

[5分]

- ☒ 在验证集性能不再提升时停止训练 (正确答案)
 - ☐ 随机丢弃神经元
 - ☐ 对权重施加惩罚
 - ☐ 增加训练数据
-

* 8. 在Dropout中, 测试阶段需要对输出做什么调整?

[5分]

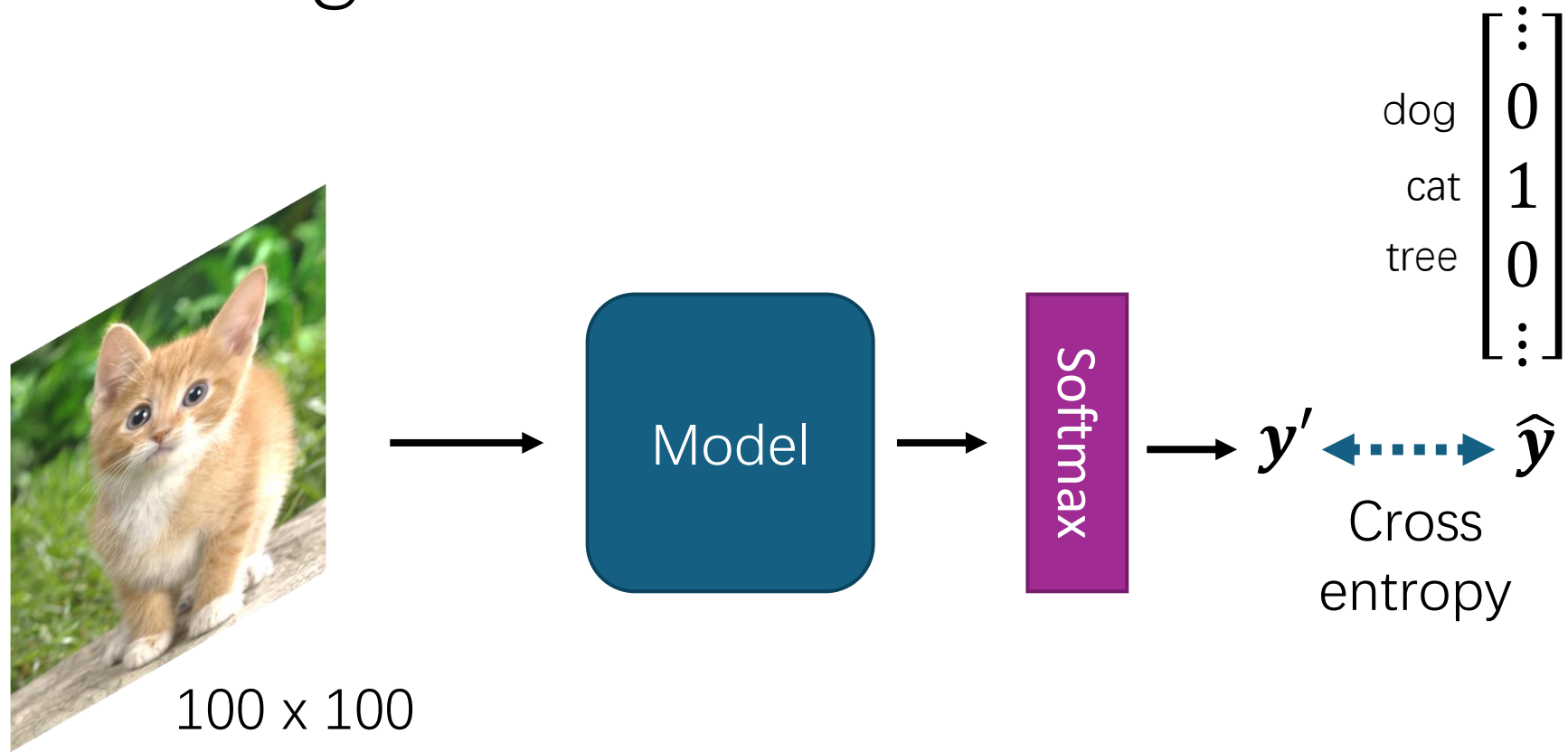
- ☐ 乘以丢弃概率
- ☐ 除以丢弃概率
- ☒ 乘以保留概率 (正确答案)
- ☐ 无需调整

Outlines

- CNN (Convolutional Neural Network)

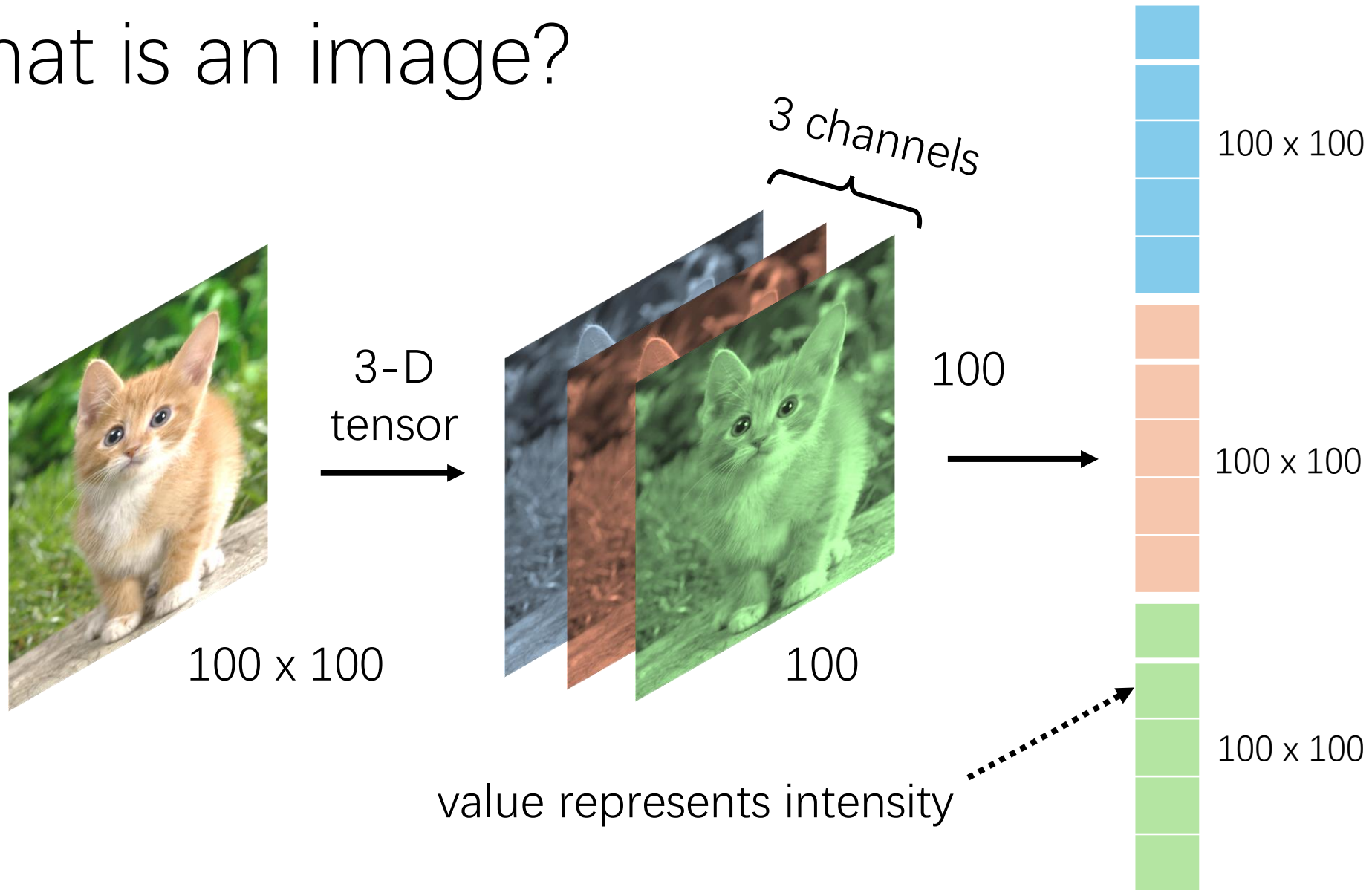
- LeNet

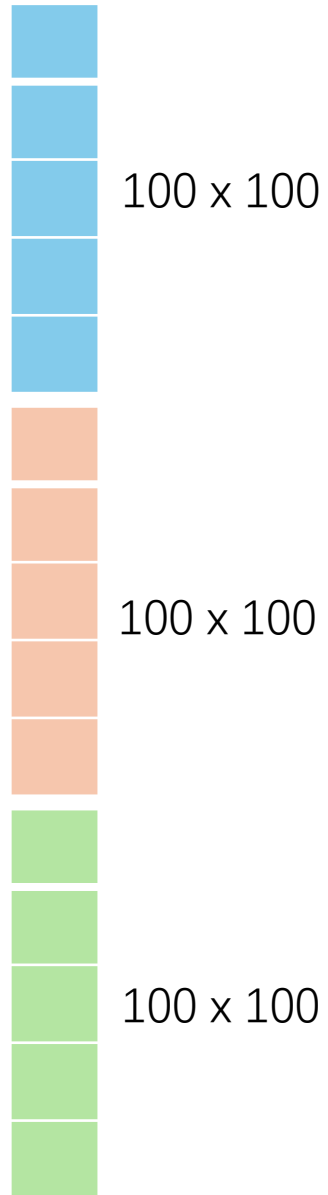
Recall: Image classification



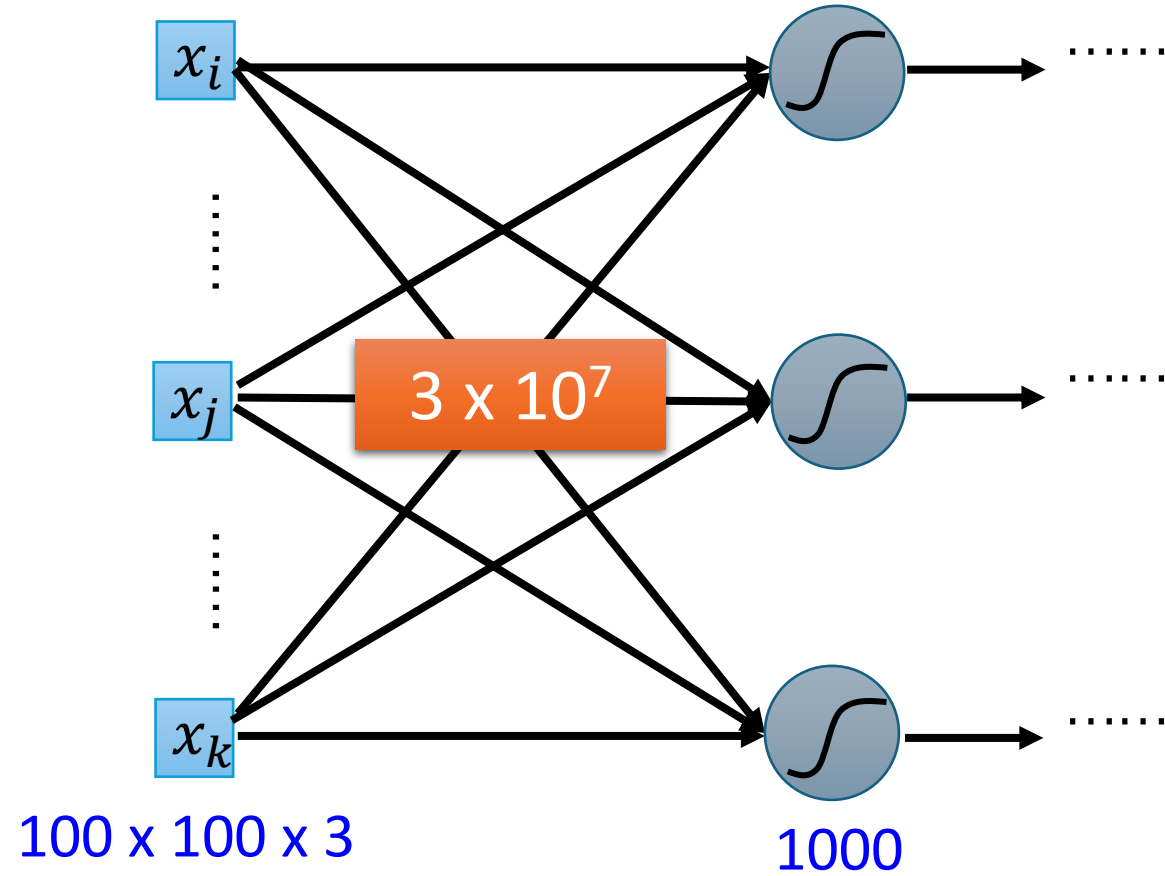
(All the images to be classified have the same size.)

What is an image?





Fully Connected Network

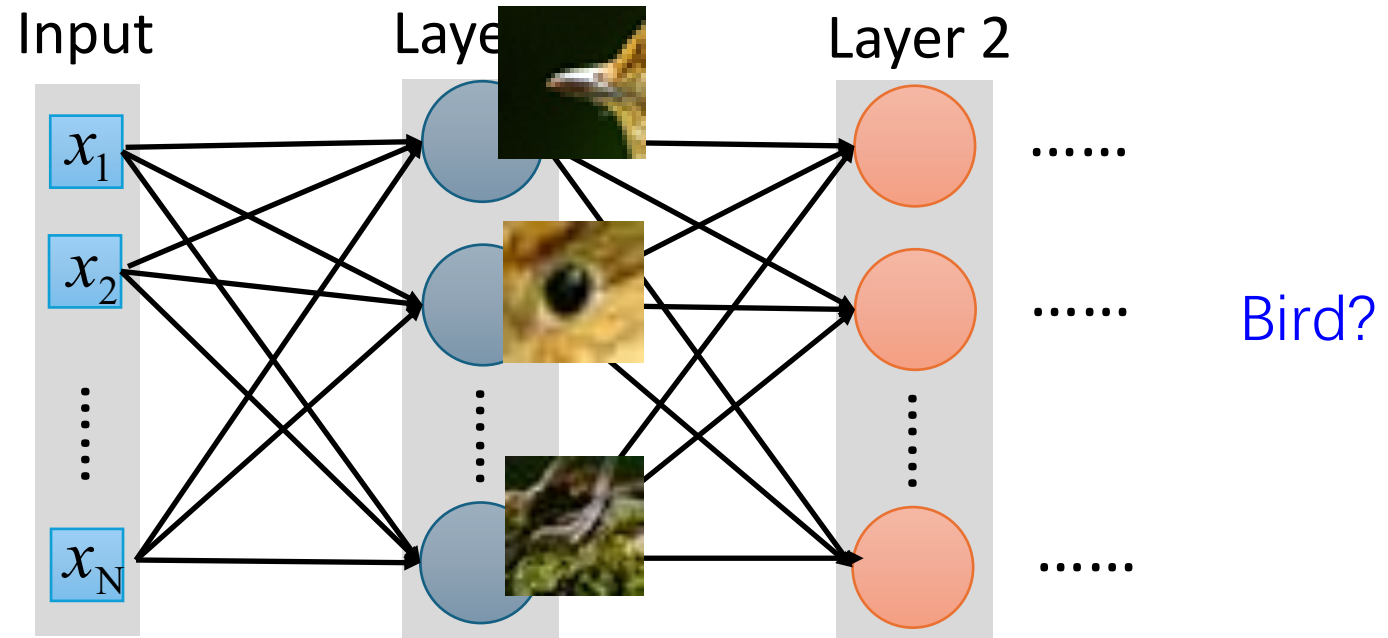


Do we really need *fully connected*
in image processing?

Observation 1

Perhaps Identifying some critical patterns is good enough.

Need to see the whole image?



Perhaps human also identify birds in a similar way ... ☺



<https://www.dcard.tw/f/funny/p/233833012>



國立中央大學

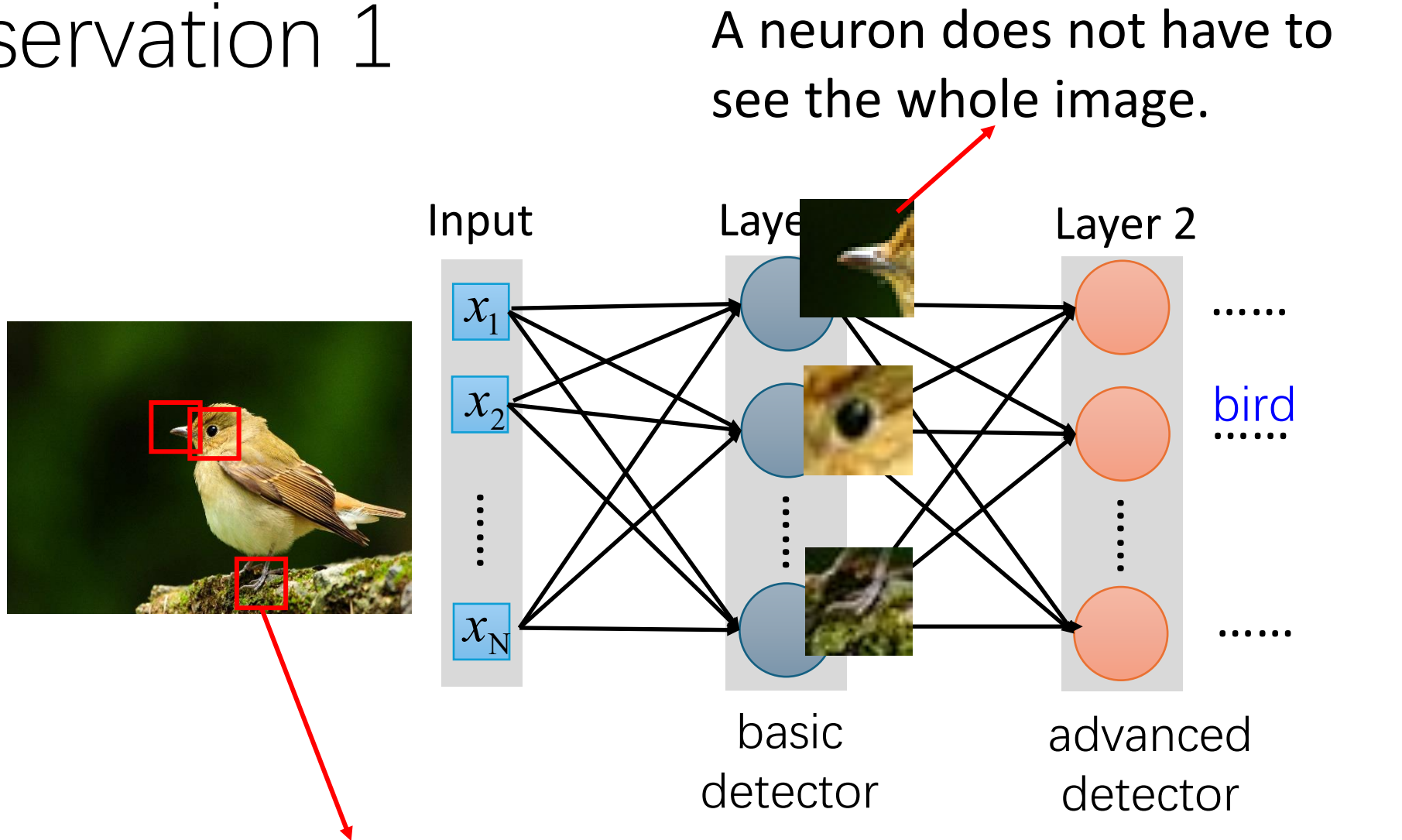
♥ 74 ↓



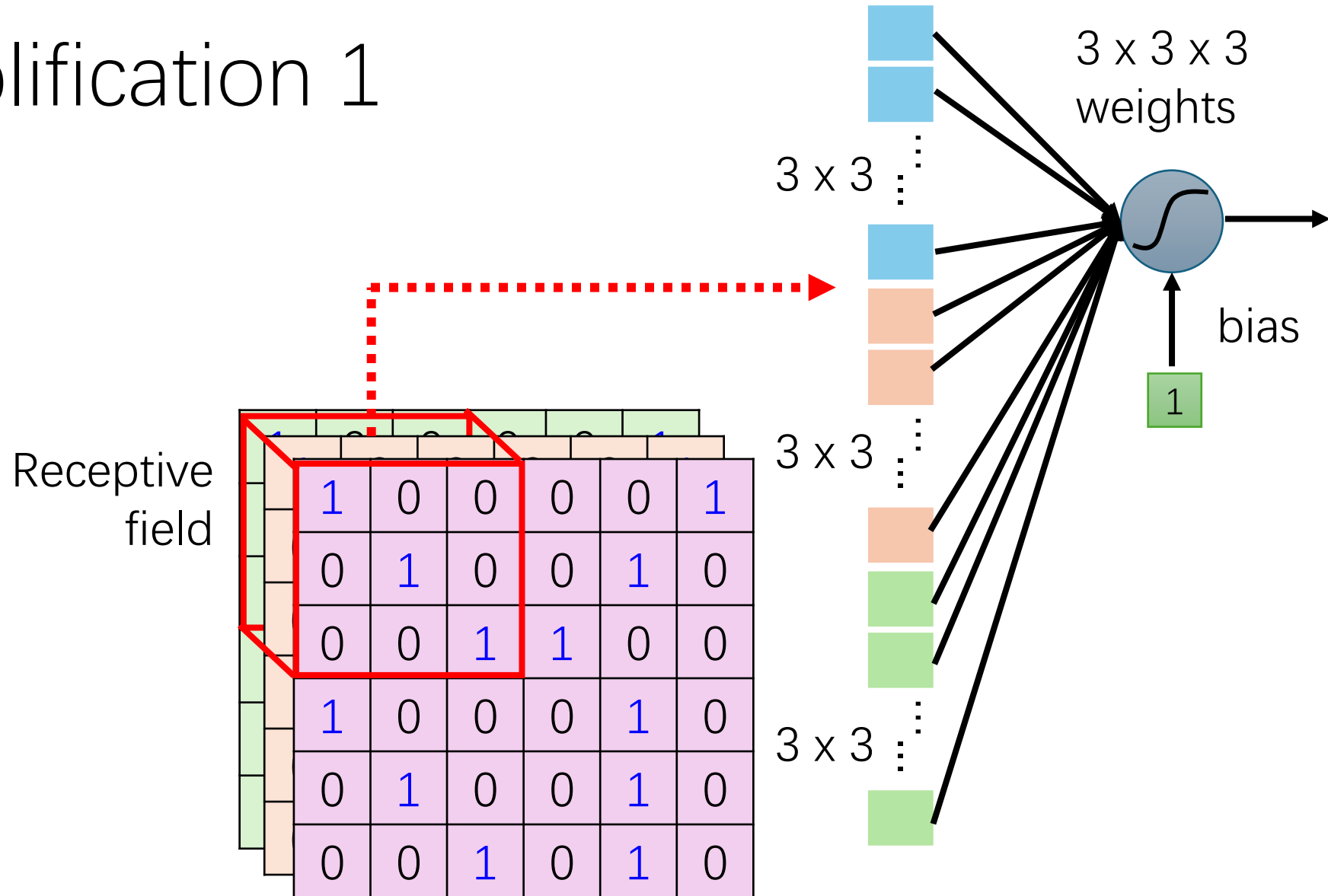
?????

B4 · 2020 年 6 月 8 日 20:37 回覆 分享

Observation 1

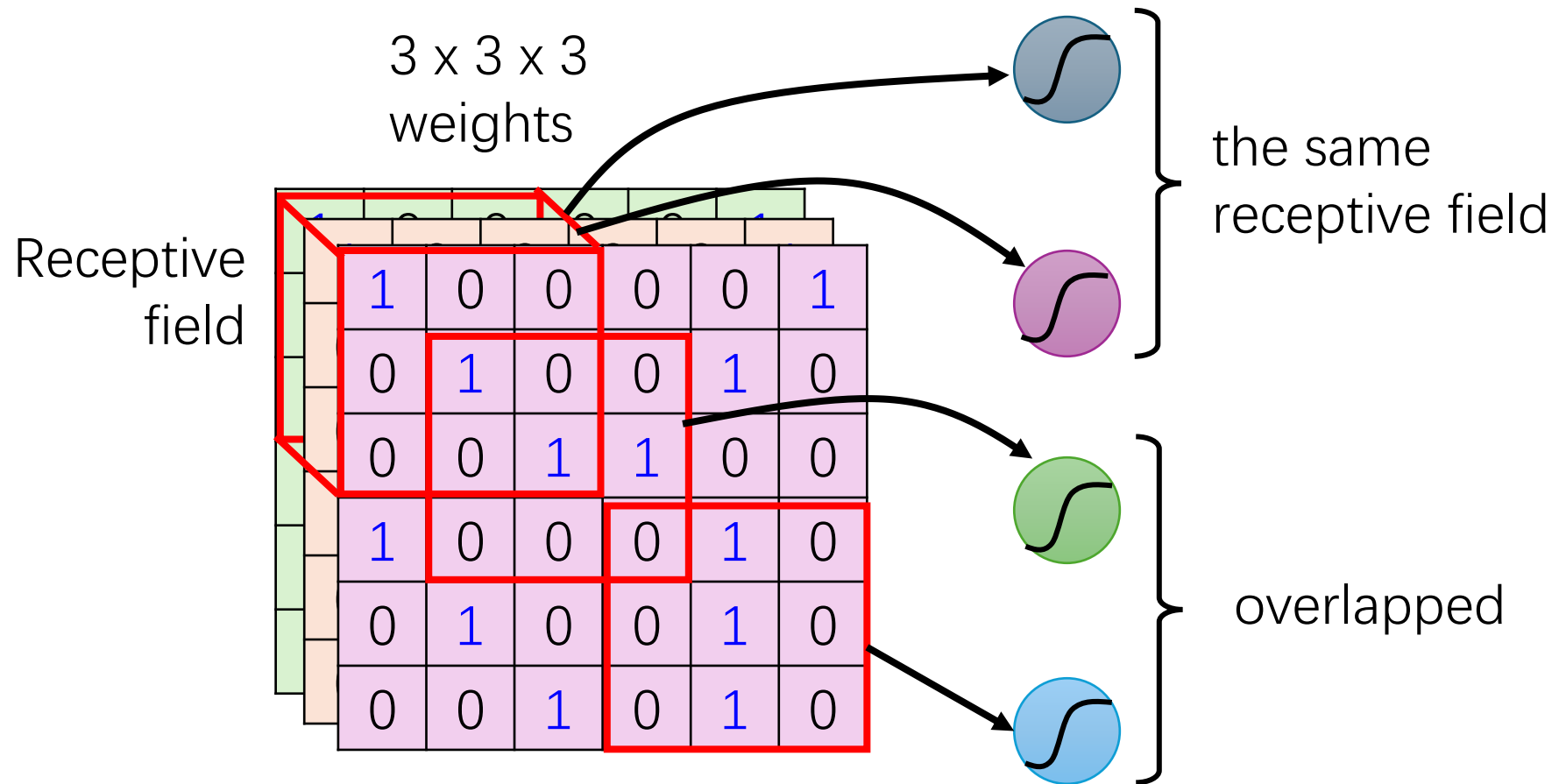


Simplification 1



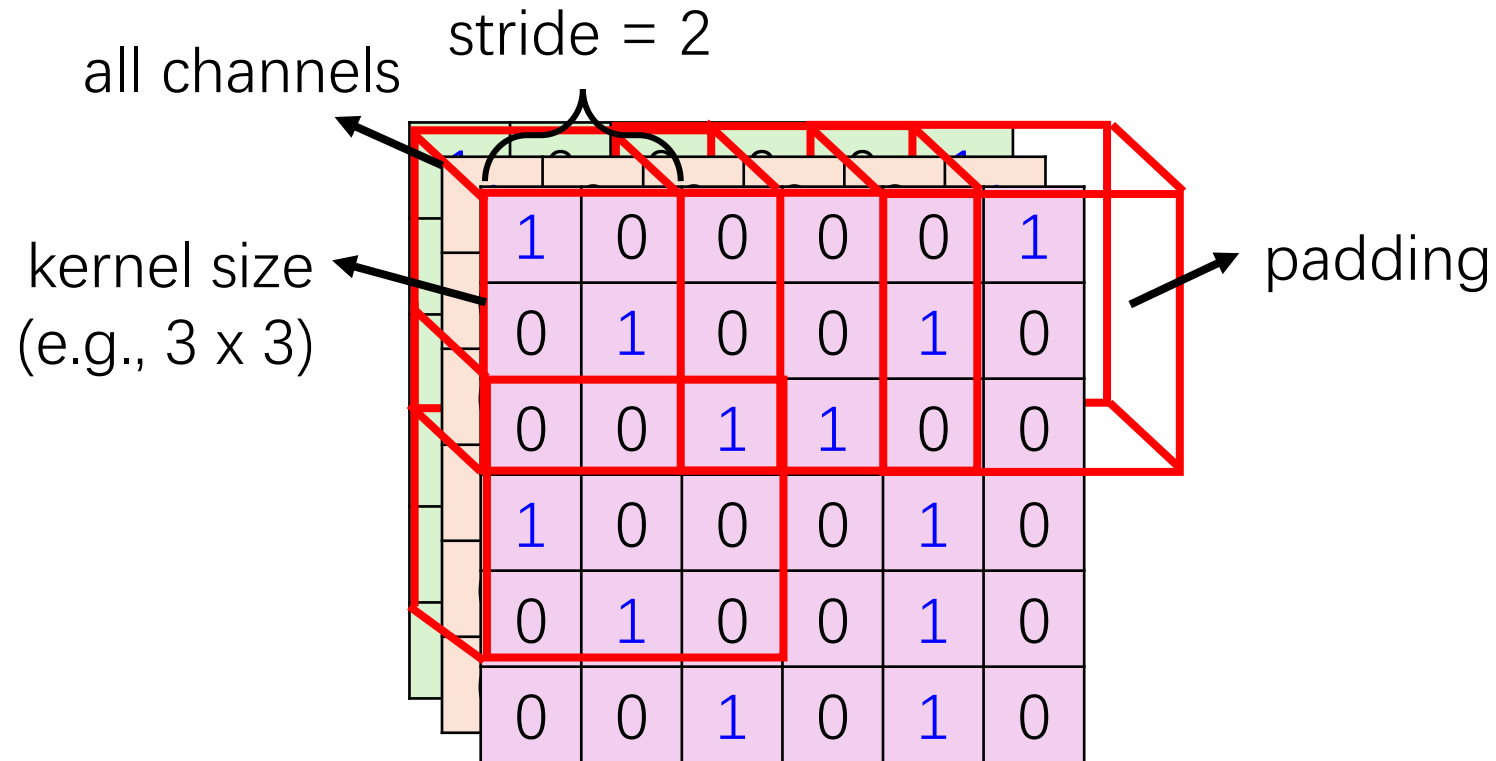
Simplification 1

- Can different neurons have different sizes of receptive field?
- Cover only some channels?
- Not square receptive field?



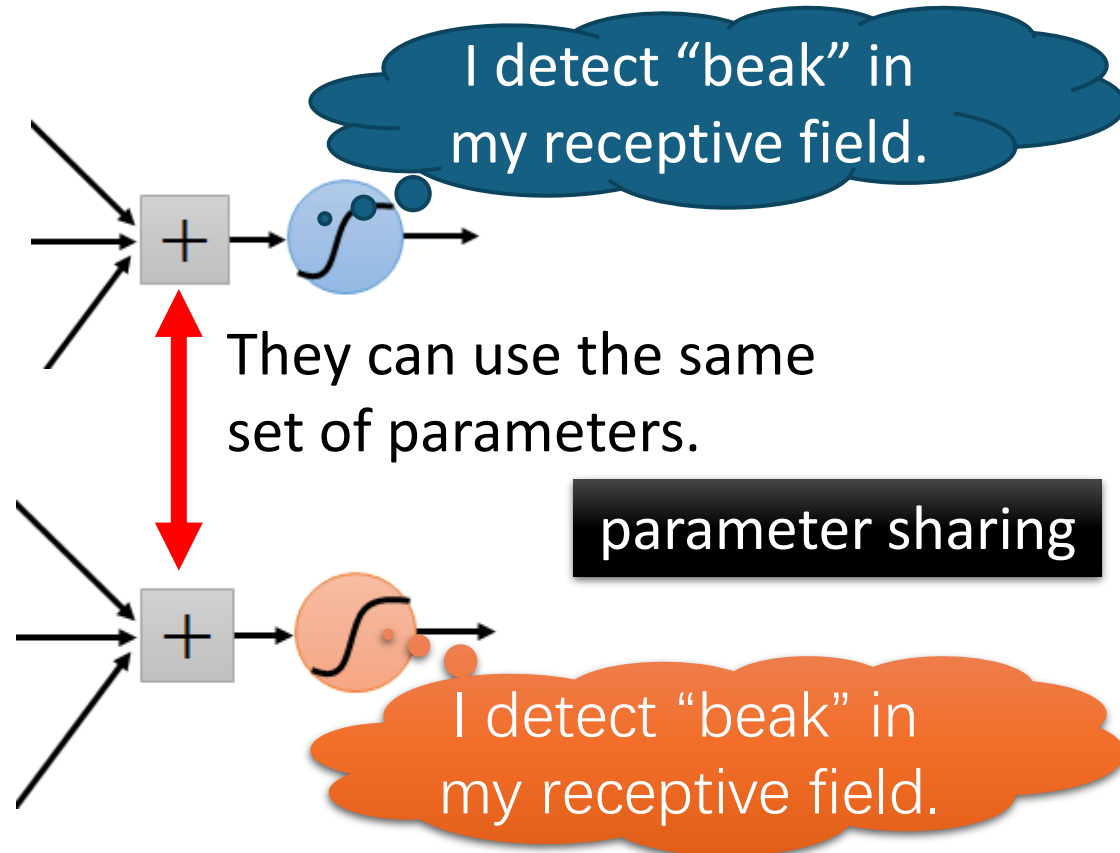
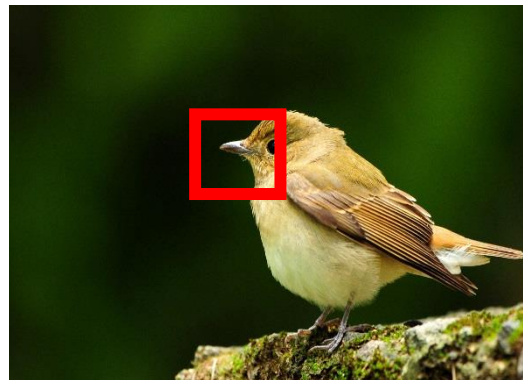
Simplification 1 – Typical Setting

Each receptive field has a set of neurons (e.g., 64 neurons).

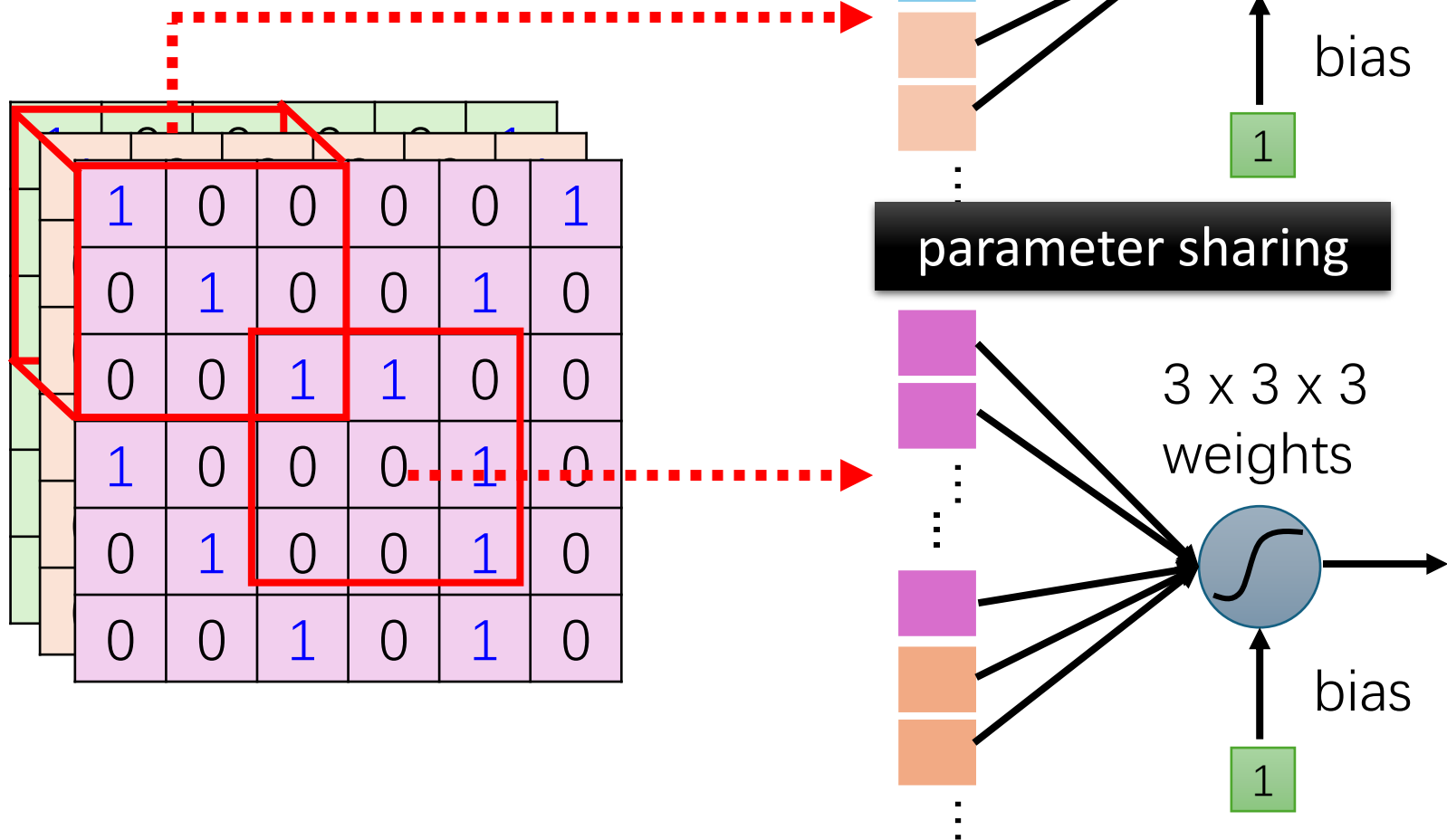


Observation 2

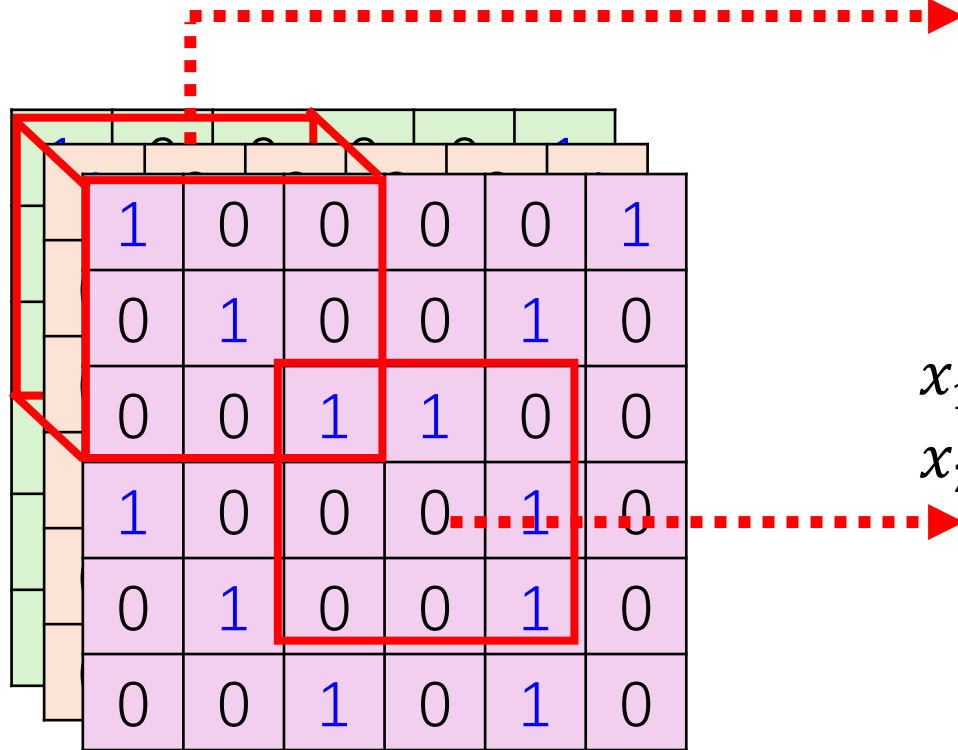
- The same patterns appear in different regions.



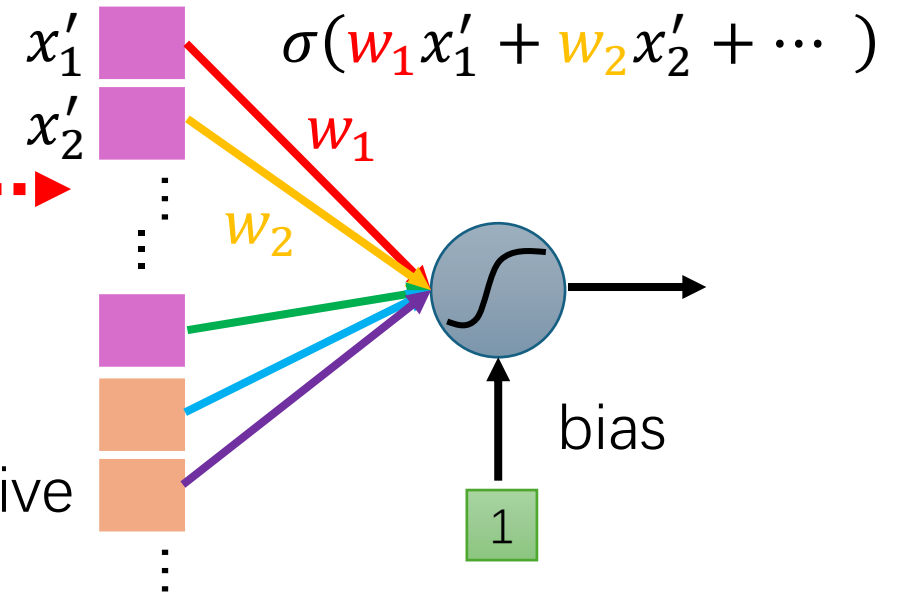
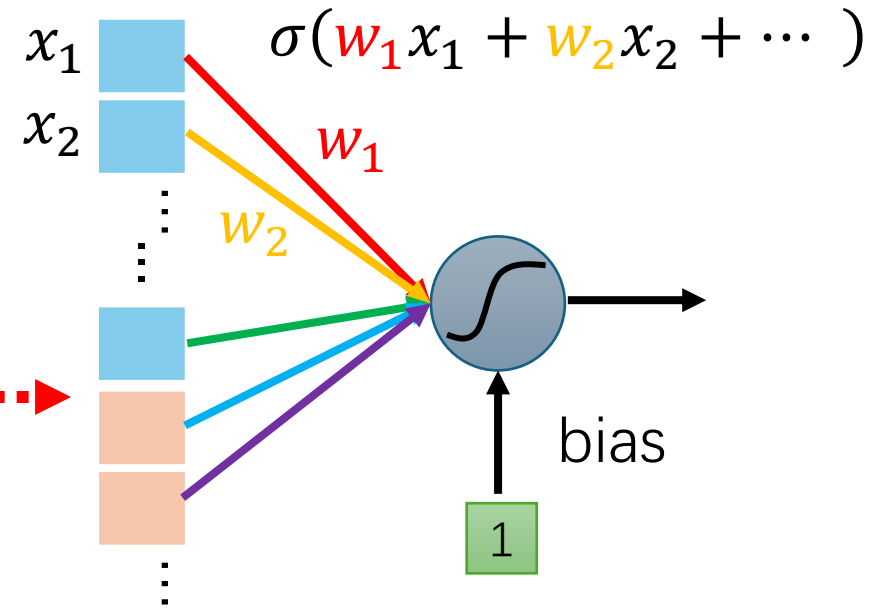
Simplification 2



Simplification 2

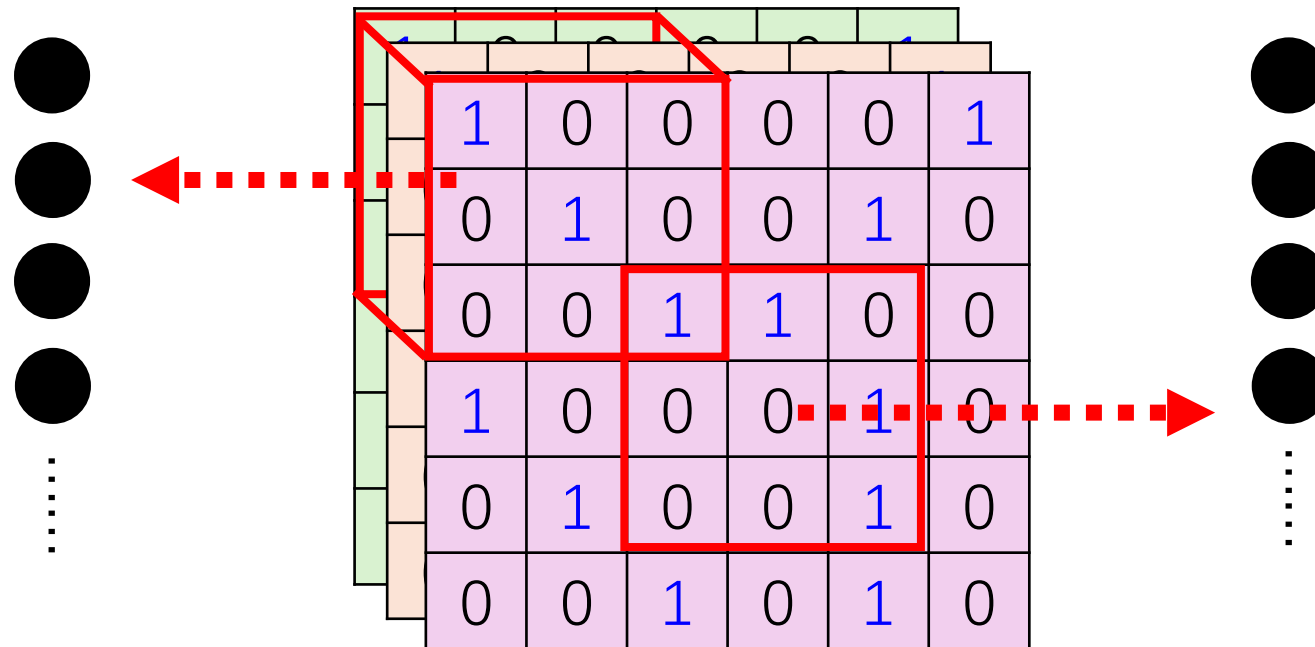


Two neurons with the same receptive field would not share parameters.



Simplification 2 – Typical Setting

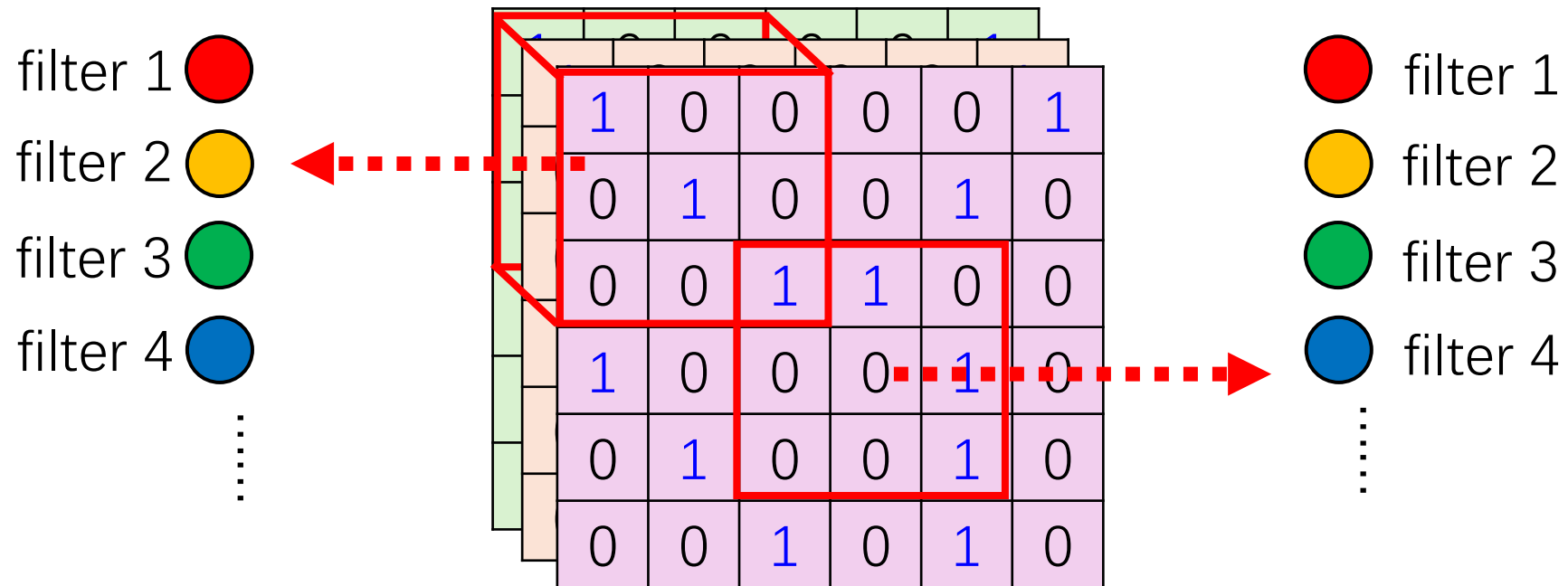
Each receptive field has a set of neurons (e.g., 64 neurons).



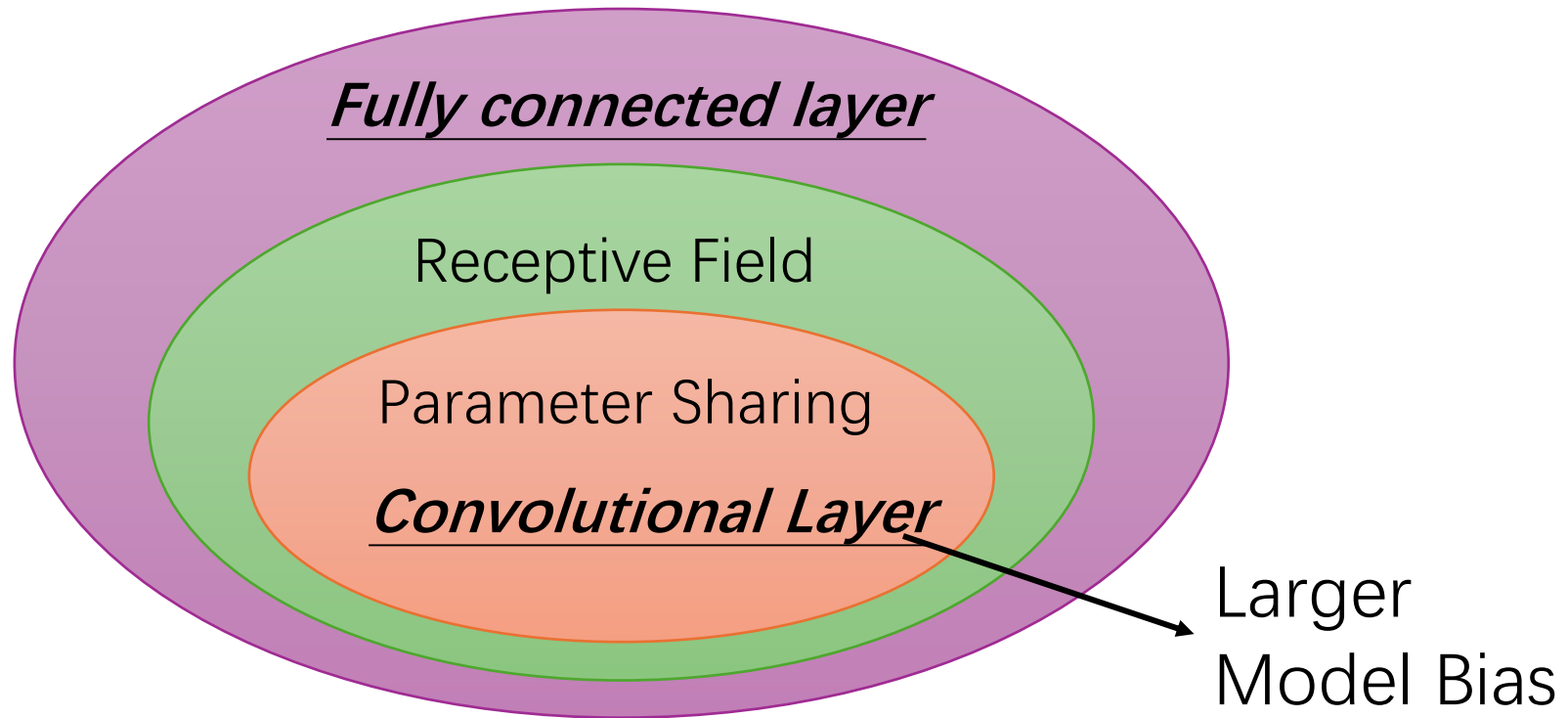
Simplification 2 – Typical Setting

Each receptive field has a set of neurons (e.g., 64 neurons).

Each receptive field has the neurons with the same set of parameters.



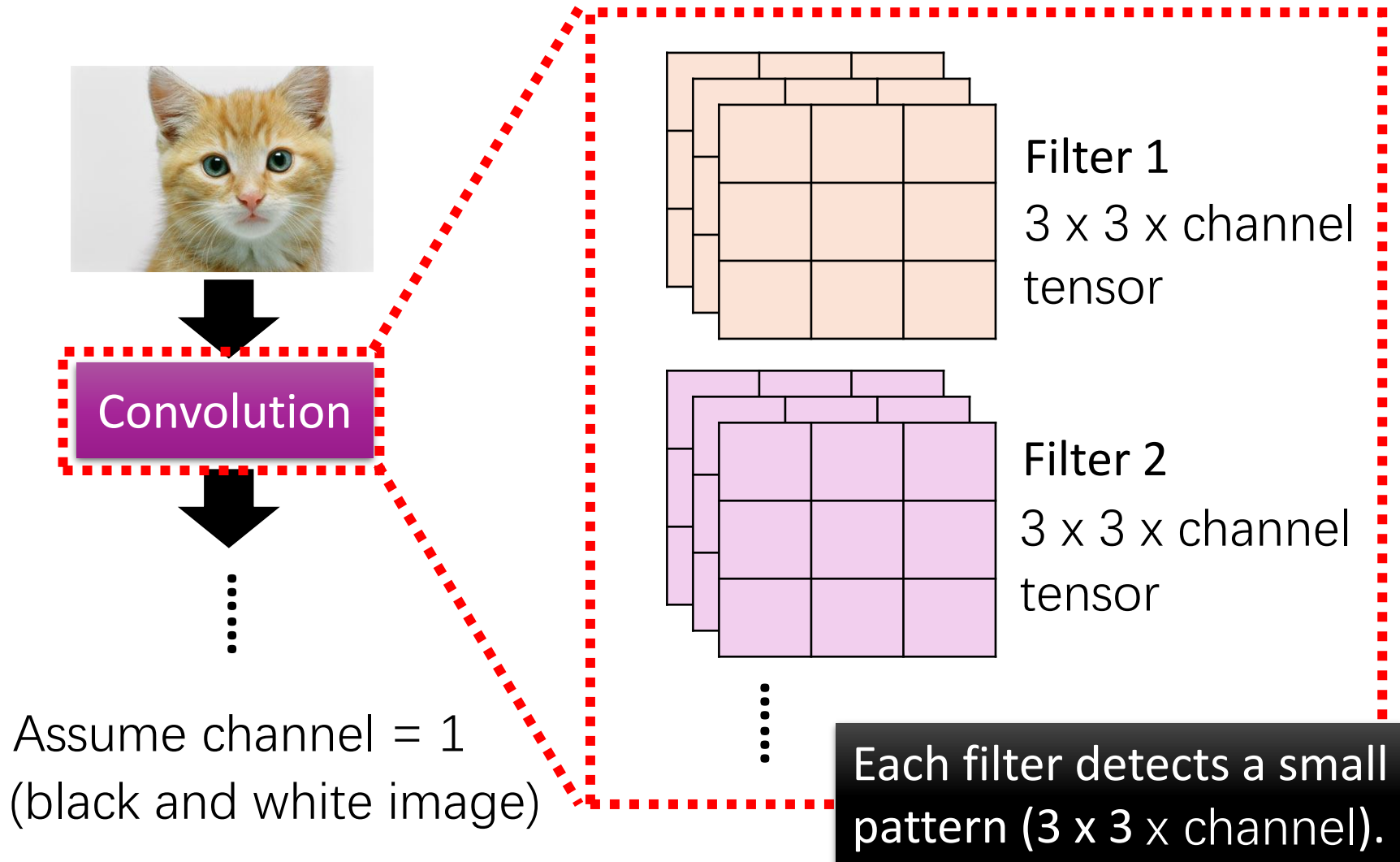
Benefit of Convolutional Layer



- Some patterns are much smaller than the whole image.
- The same patterns appear in different regions.

Convolutional Layer

Another story based on filter ☺



Convolutional Layer

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

Convolutional Layer

stride=1

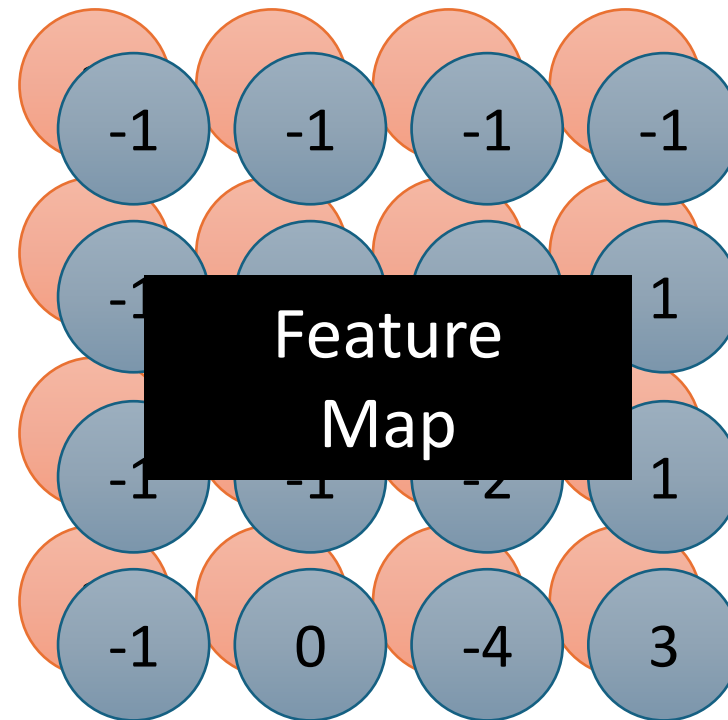
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

Do the same process for every filter



4 x 4 image

Convolutional Layer

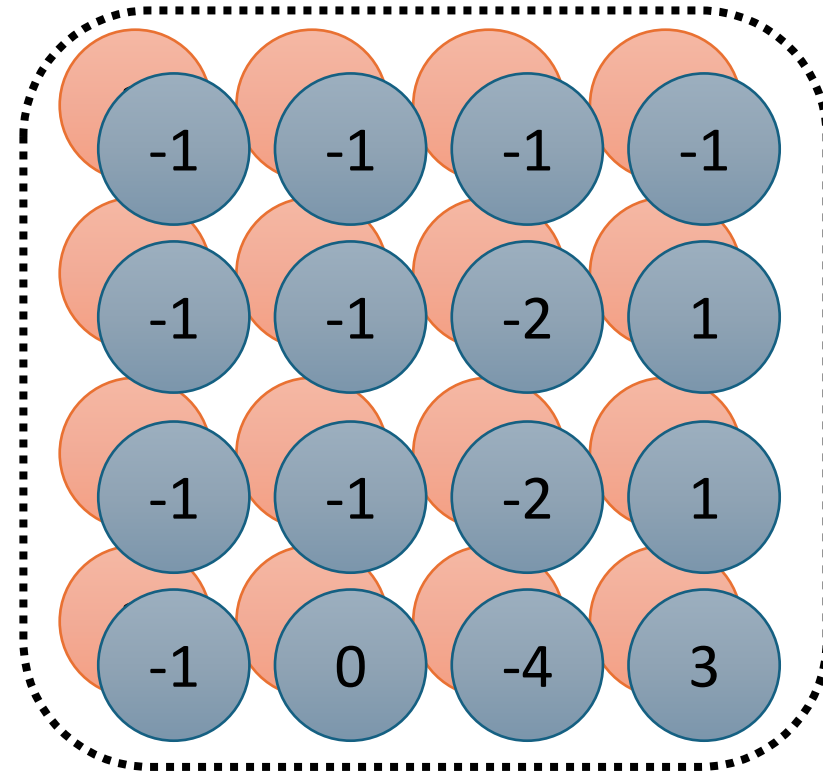


64
filters

Convolution

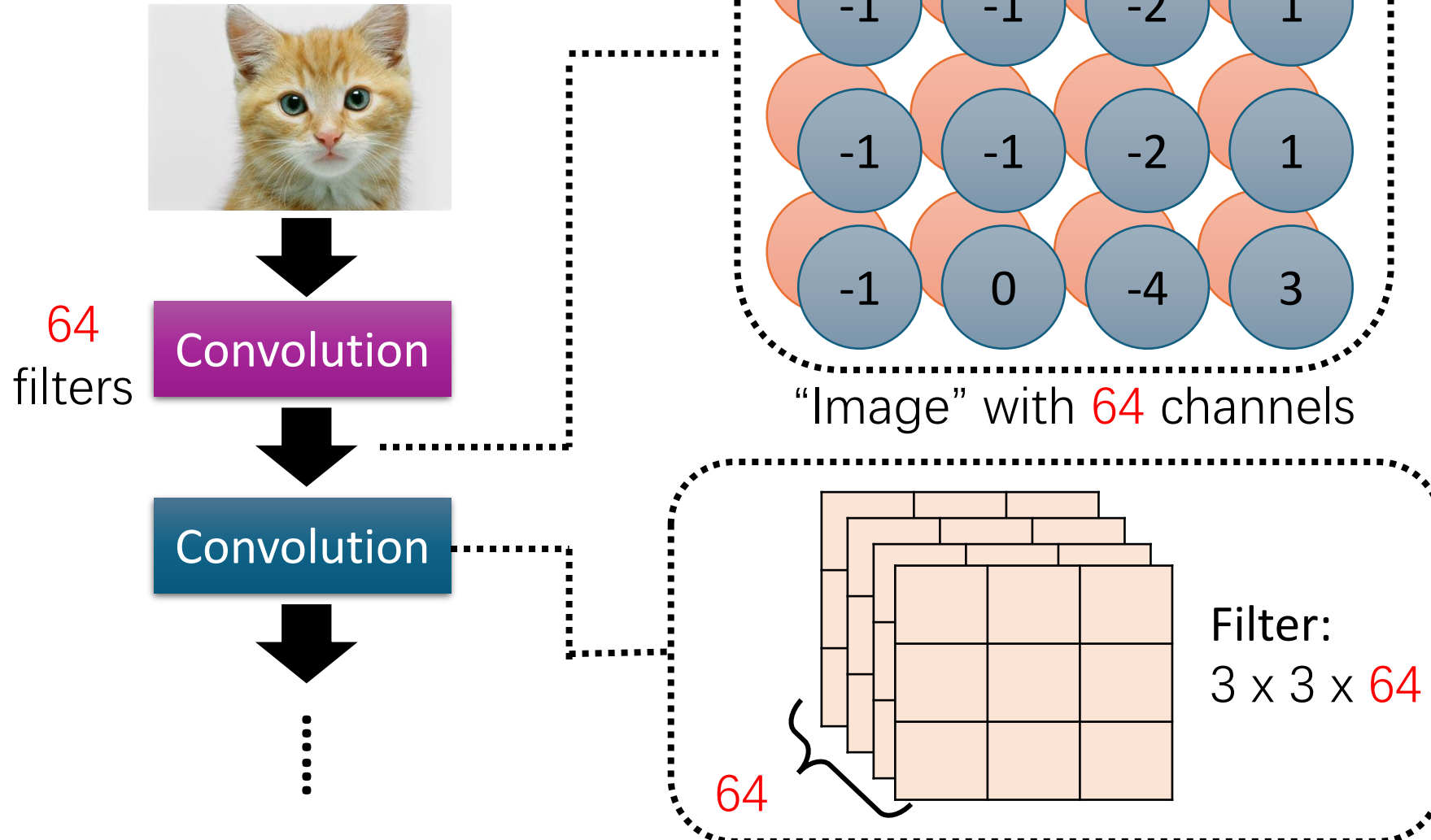
Convolution

⋮

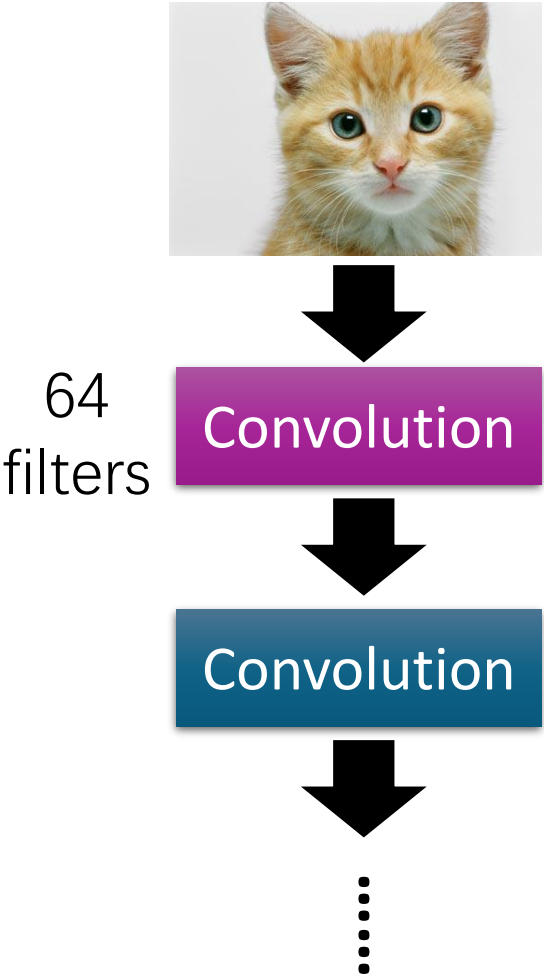


"Image" with 64 channels

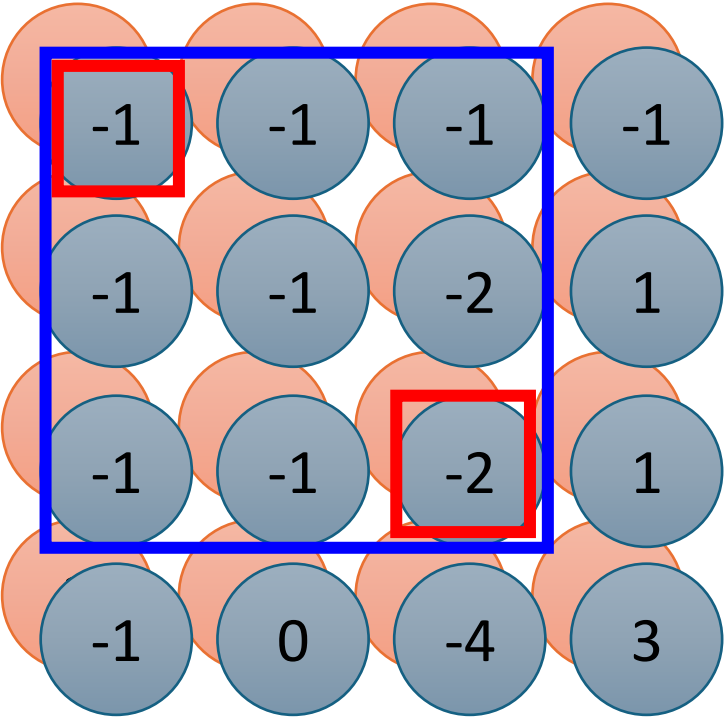
Multiple Convolutional Layers



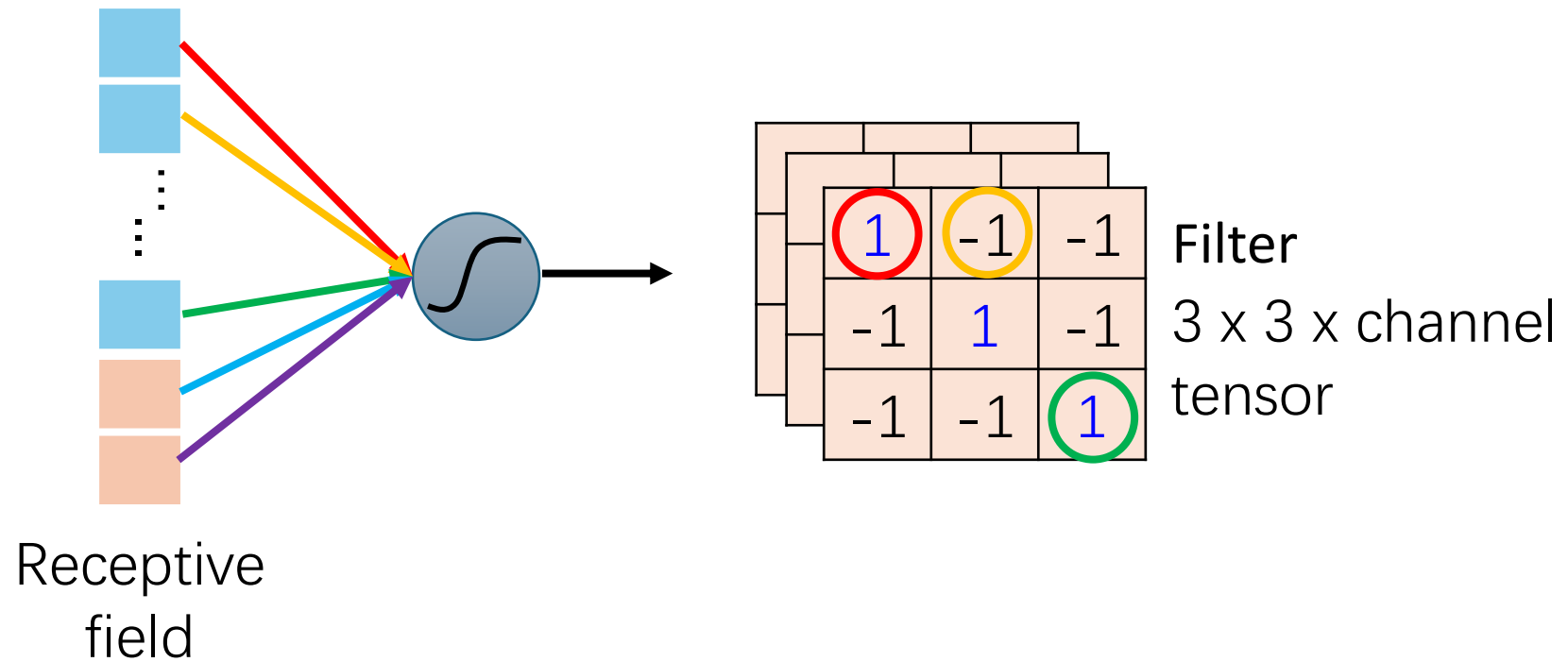
Multiple Convolutional Layers



1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0



Comparison of Two Stories



Convolutional Layer

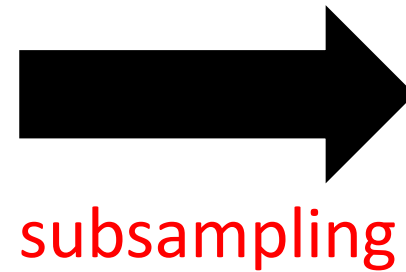
<u><i>Neuron Version Story</i></u>	<u><i>Filter Version Story</i></u>
Each neuron only considers a receptive field.	The set parameters of each neuron is a filter.
Some neurons use the same set of parameters.	Each filter convolves over the input image.

They are the same story.

Pooling

- Subsampling the pixels will not change the object

bird



bird



Pooling – Max Pooling

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

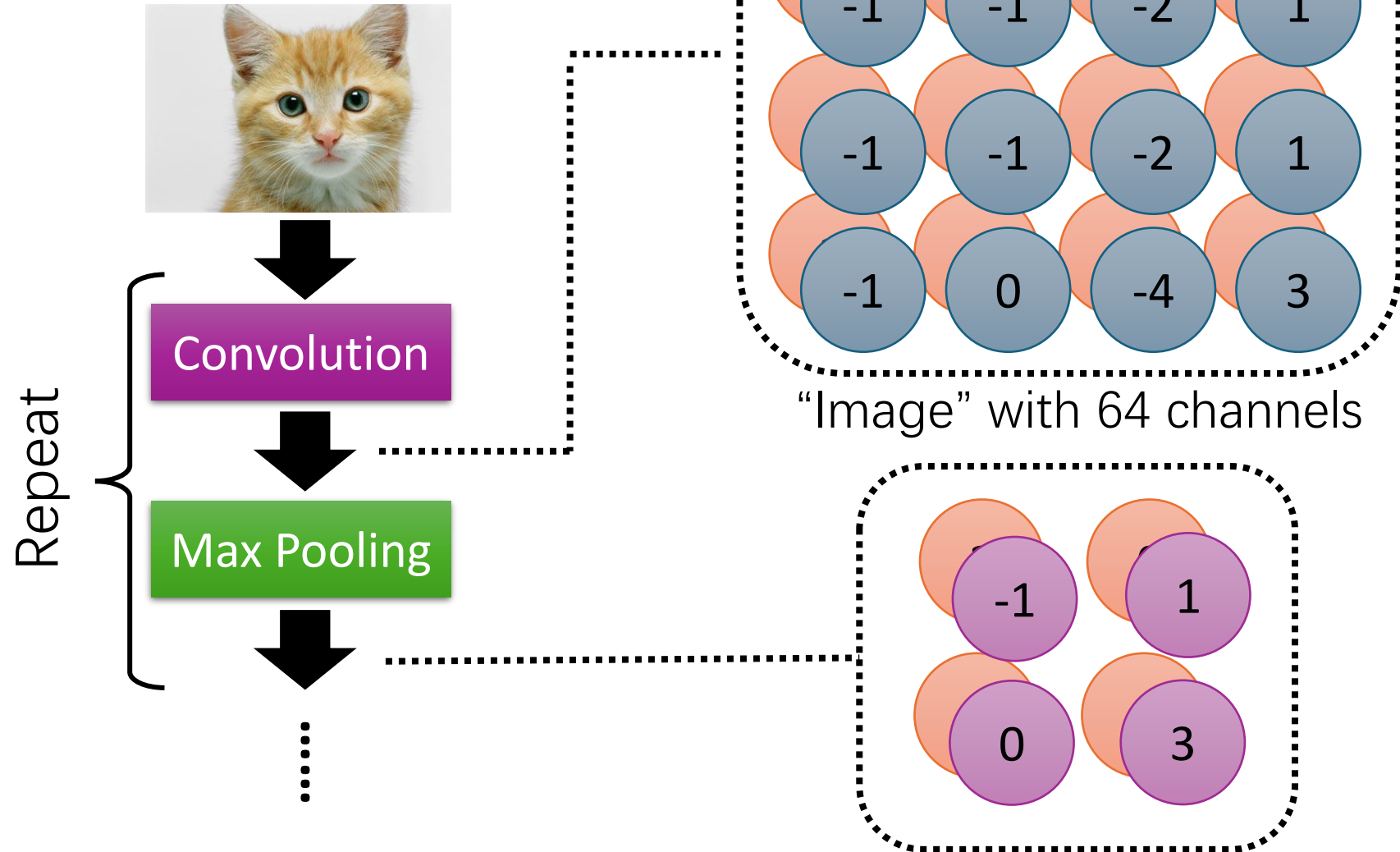
-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

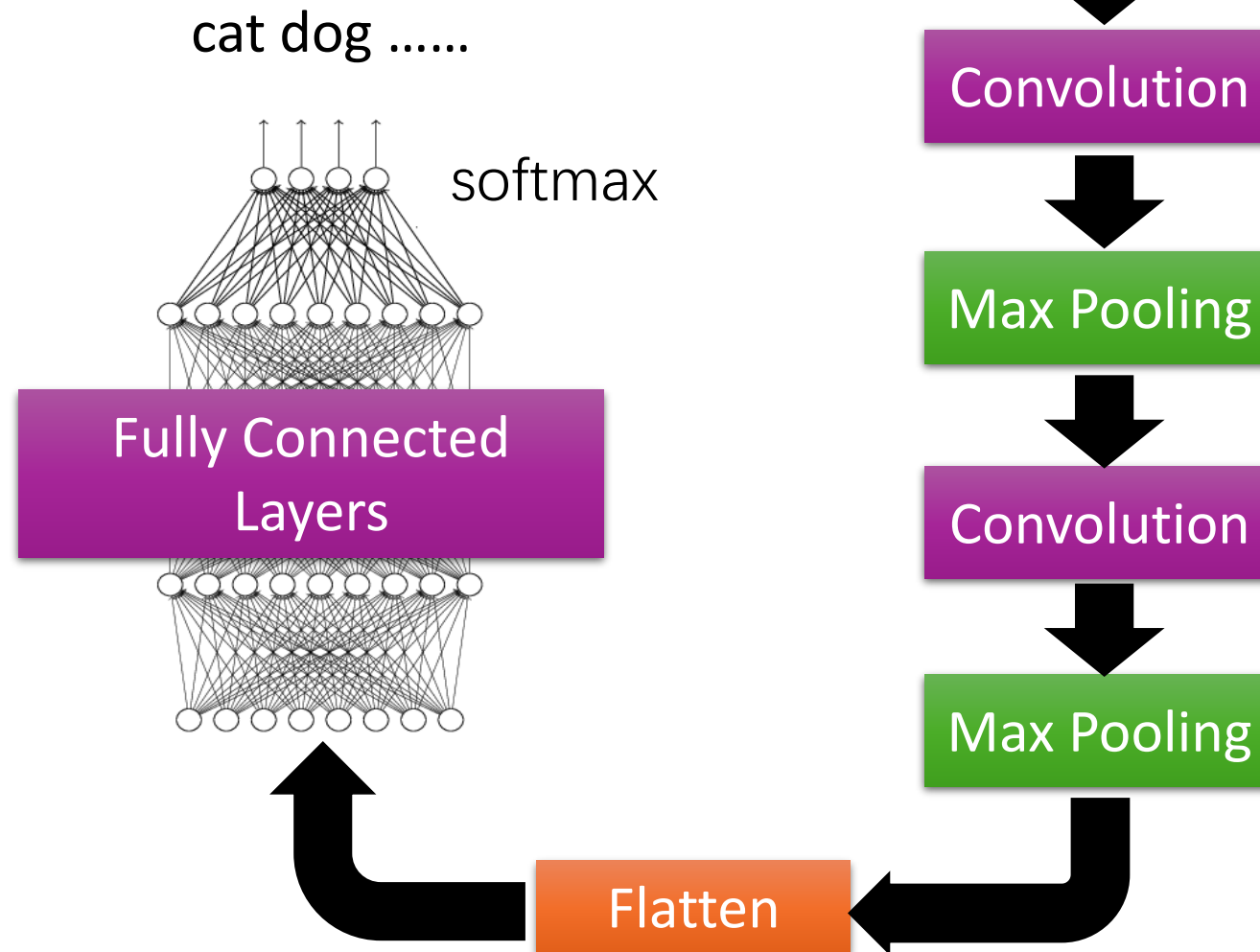
3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

-1	-1	-1	-1
-1	-1	-2	1
-1	-1	-2	1
-1	0	-4	3

Convolutional Layers + Pooling



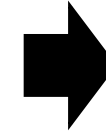
The whole CNN



Application: Playing Go



Network



Next move
(19 x 19
positions)

19 x 19 classes

48 channels in
Alpha Go

Black: 1
white: -1
none: 0

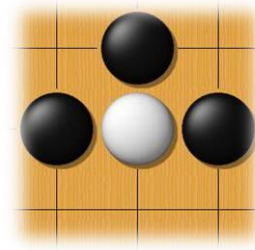
Fully-connected feedforward
network can be used

But CNN performs much better.

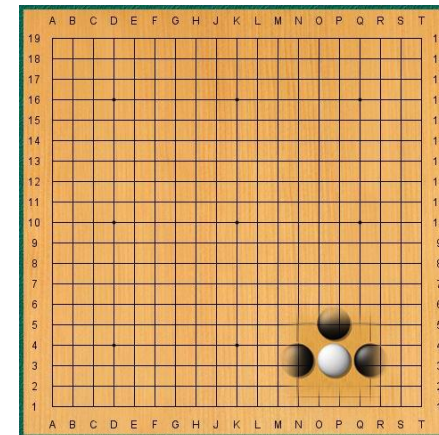
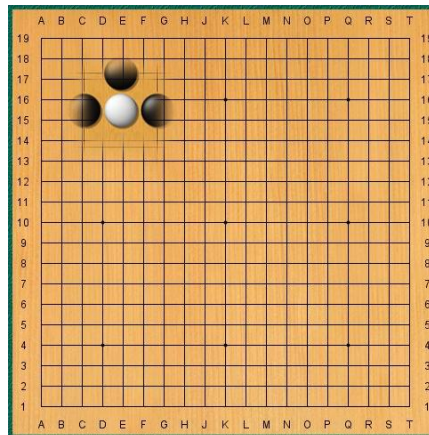
Why CNN for Go playing?

- Some patterns are much smaller than the whole image

Alpha Go uses 5 x 5 for first layer



- The same patterns appear in different regions.



Why CNN for Go playing?

- Subsampling the pixels will not change the object



Max Pooling

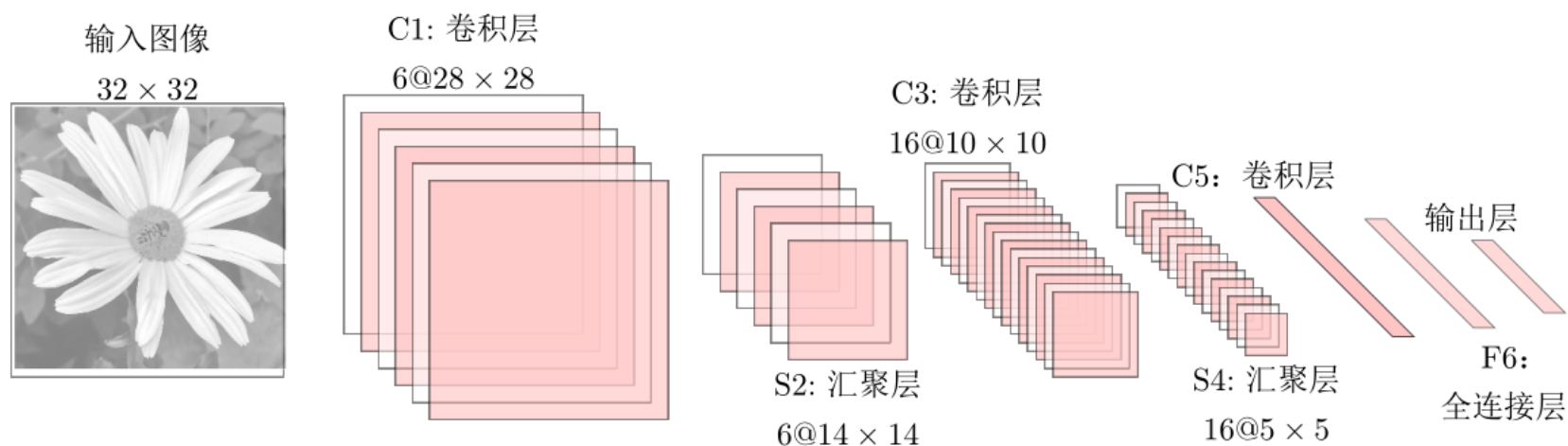
How to explain this???

Neural network architecture. The input to the policy network is a $19 \times 19 \times 48$ image stack consisting of 48 feature planes. The first hidden layer zero pads the input into a 23×23 image, then convolves k filters of kernel size 5×5 with stride 1 with the input image and applies a rectifier nonlinearity. Each of the subsequent hidden layers 2 to 12 zero pads the respective previous hidden layer into a 21×21 image, then convolves k filters of kernel size 3×3 with stride 1, again followed by a rectifier nonlinearity. The final layer convolves 1 filter of kernel size 1×1 with stride 1, with a different bias for each position, and applies a softmax function. The match version of AlphaGo used $k = 192$ filters; Fig. 2b and Extended Data Table 3 additionally show the results of training with $k = 128, 256$ and 384 filters.

Alpha Go does not use Max Pooling

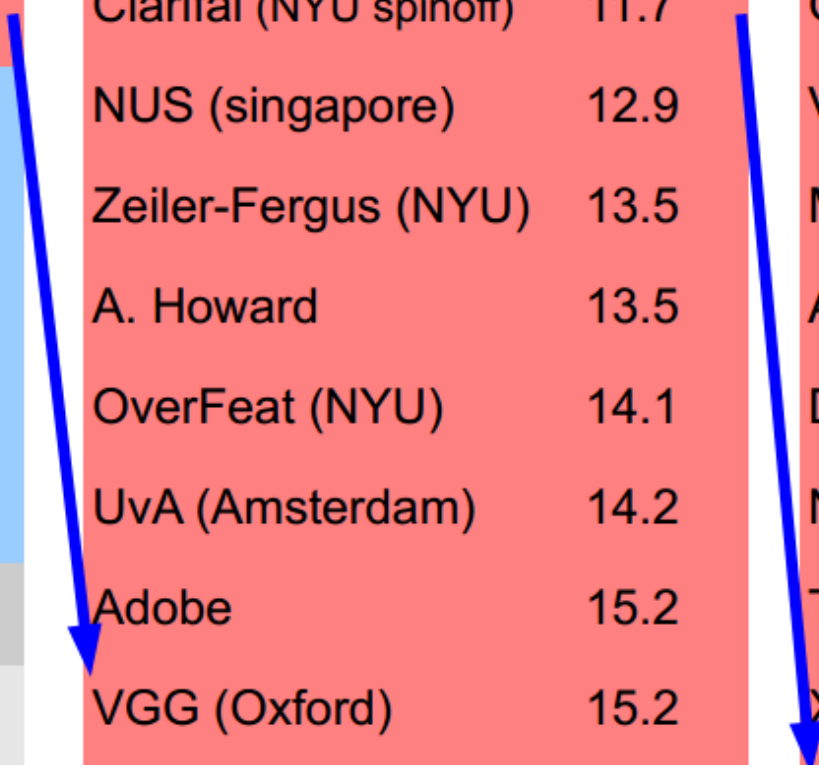
LeNet-5

- LeNet-5 是一个非常成功的神经网络模型。
 - 基于 LeNet-5 的手写数字识别系统在 90 年代被美国很多银行使用，用来识别支票上面的手写数字。
 - LeNet-5 共有 7 层。



Large Scale Visual Recognition Challenge

2012 Teams	%error	2013 Teams	%error	2014 Teams	%error
Supervision (Toronto)	15.3	Clarifai (NYU spinoff)	11.7	GoogLeNet	6.6
ISI (Tokyo)	26.1	NUS (singapore)	12.9	VGG (Oxford)	7.3
VGG (Oxford)	26.9	Zeiler-Fergus (NYU)	13.5	MSRA	8.0
XRCE/INRIA	27.0	A. Howard	13.5	A. Howard	8.1
UvA (Amsterdam)	29.6	OverFeat (NYU)	14.1	DeeperVision	9.5
INRIA/LEAR	33.4	UvA (Amsterdam)	14.2	NUS-BST	9.7
		Adobe	15.2	TTIC-ECP	10.2
		VGG (Oxford)	15.2	XYZ	11.2
		VGG (Oxford)	23.0	UvA	12.1



Keywords of CNN

- Receptive Field 感受野 （局部连接）
- Parameter share 权重共享
- Max Pooling 最大池化
- Stride 步长
- Kernel size 卷积核大小
- Padding 填充

Survey



Next course

- **ResNet + review**

Chapter 7, 8 in 《dive into deep learning》

Chapter 10 in 《Deep learning》